

DEPT. OF ELECTRICAL & ELECTRONICS ENGINEERING
SRM UNIVERSITY, Kattankulathur – 603203.

Experiment No : 2a & 2b

Title of Experiment : **Measurement of L and C using Bridge Circuits**

Name of the candidate :

Register Number :

Date of Experiment :

Date of Submission :

S.No	Marks split up	Maximum Marks	Marks Obtained
1	Prerequisite Knowledge	5	
2	Conduction of Experiment	10	
3	Observation/Inference	10	
4	Post Practicum Knowledge	5	
	Total	30	

Instructor Signature

Experiment No. 2a	MEASUREMENT OF INDUCTANCE USING
Date:	MAXWELL'S BRIDGE

AIM:

To design and determine the self-inductance of a given coil using Maxwell's Inductance Bridge.

APPARATUS REQUIRED:

Sl.No.	Apparatus	Range	Quantity
1	Maxwell's bridge kit	--	1
2	Oscillator	--	1
3	Decade Inductance box	--	1
4	Head phone	--	1
5	Patch card	--	-

THEORY:

Maxwell's Inductance Bridge measures an inductance comparison with a variable standard self-inductance. The connection for balanced condition is shown in the figure below

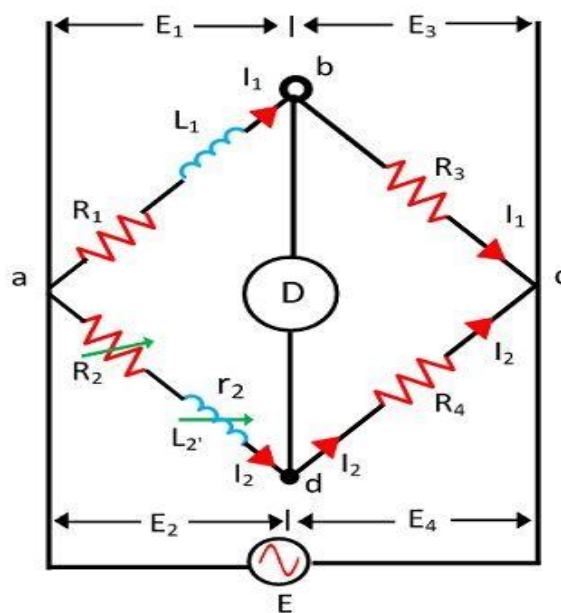


Figure Maxwell's Inductance Bridge Circuit

Let, L_1 – unknown inductance of resistance R_1 .
 L_2 – Variable inductance of fixed resistance r_1 .
 R_2 – variable resistance connected in series with inductor L_2 .
 R_3, R_4 – known non-inductance resistance

At balance,

$$L_1 = \frac{R_3}{R_4} L_2$$

$$R_1 = \frac{R_3}{R_4} (R_2 + r_2)$$

FORMULA USED:

Unknown Inductance (L_1)

$$L_1 = \frac{R_3}{R_4} L_2$$

$$R_1 = \frac{R_3}{R_4} (R_2 + r_2)$$

PROCEDURE:

1. Connect the circuit as shown in the figure.
2. Connect the unknown inductance in L_1 .
3. Connect the Head phone between ground and output of imbalance amplifier.
4. Vary R_2 , from minimum position, in clockwise direction.
5. If the selection of R_2 is correct the balance point can be obtained at minimum position.
6. Vary R_2 for fine balance adjustment.

TABULATION:

S.No.	Resistance, R_2	Resistance, R_3	Unknown Inductance (L_1)	Given value of inductance, L
Unit	$k\Omega$	$k\Omega$	mH	H
1				
2				
3				

RESULT:

The value of the inductance of the given coil is measured using Maxwell's Inductance Bridge using the formula derived from the theoretical concept and the value of self-inductance is -----H

Experiment No. 2b	MEASUREMENT OF CAPACITANCE USING SCHERING BRIDGE
Date:	

AIM:

To design and determine the unknown capacitance and dissipation factor using Schering's bridge.

APPARATUS REQUIRED:

Sl.No.	Apparatus	Range	Quantity
1	Schering's bridge kit	--	1
2	Oscillator	--	1
3	Decade Capacitance box	--	1
4	Head phone	--	1
5	Unknown Capacitance	--	1
6	Patch card	--	-

THEORY:

Schering's Bridge is a widely used bridge circuit for the measurement of capacitance and dissipation factor and the same is shown in the below Figure.

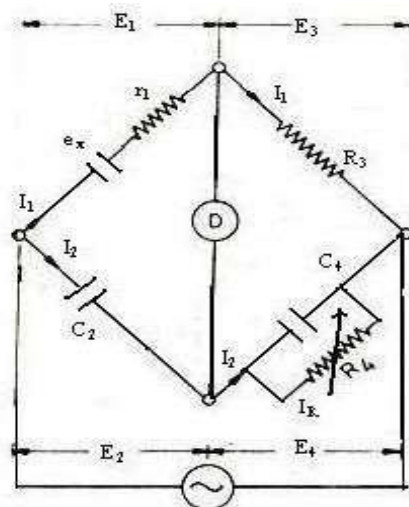


Figure Schering's Bridge Circuit

The measurement at balance is as given below.

$$\begin{aligned} (r_1 + 1 / j\omega C_x) (R_4 / (1 + j\omega C_4 R_4)) &= 1 / j\omega C_2 \times R_3 \\ r_1 R_4 - jR_4 / \omega C_x &= -jR_3 / \omega C_2 + R_3 R_4 C_4 / C_2 \end{aligned}$$

Equating the real and imaginary parts,

$$\begin{aligned} R_1 &= R_3 C_4 / C_2 \\ C_x &= C_2 (R_4 / R_3) \end{aligned}$$

FORMULA USED:

$$\begin{aligned} C_x &= C_2 (R_4 / R_3) \\ \text{Where, } C_2 &= \text{Standard capacitor} \\ R_3, R_4 &= \text{Non-inductive} \end{aligned}$$

Resistance Dissipation factor,

$$D_1 = \tan \delta = \omega C_1 r_1 = \omega C_4 R_4$$

Where, C_x – Unknown capacitance

C_4 – variable Capacitor

R_3, R_4 – Non inductive resistance

PROCEDURE:

- The connections to be made as per the circuit diagram (Tainer kit connections) and the unknown capacitance is connected in the terminals (C_x)
- Initially the resistance R_3 is kept some value and by varying the value of resistance R_4 the balanced condition is obtained.
- The balanced condition is checked with the help of head phone.
- All the values are noted down.

TABULATION:

S.No.	Resistance, R_3	Capacitance, C_2	Resistance, R_4	$L=(RQ+m(S+R))$	Given value of Capacitance, C
Unit	$k\Omega$	μF	Ω	μF	μF
1					
2					
3					
4					

RESULT:

The value of the unknown capacitance is measured using Schering's bridge using the formula derived from the theoretical concept and the value of unknown capacitance is ----- μF .